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Harnessing confinement effect and interpretable machine learning to predict alkane diffusion in zeolite catalysts

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Abstract

Diffusion within zeolites is a critical determinant of their performance in catalysis and separations. Nevertheless, mechanistic insights remain largely system-specific, and a universal framework connecting topological features to diffusion properties is still lacking. Here, we address this challenge by combining high-throughput molecular simulations across approximately 100,000 zeolitic frameworks from the database of zeolite structures with interpretable machine learning. Guided by confinement-diffusion mechanisms, we propose a set of topology-informed descriptors (including tortuosity and cross-sectional variance) and construct a highly accurate, transferable model that quantifies the contribution of each structural factor. Our findings indicate that although pore-limiting diameters primarily facilitate diffusion, channel tortuosity and cross-sectional heterogeneity act as major transport barriers. Transfer learning further demonstrates that the framework trained on methane diffusion can be efficiently extended to other small organic molecules (e.g., ethane, ethene, and methanol). This study provides a large, systematically curated dataset for zeolite diffusion and establishes a machine learning-based framework to elucidate confinement-governed mass transport, thereby accelerating the rational design of tailored zeolite materials.

Introduction

Zeolites, a class of crystalline microporous aluminosilicates, are widely employed in catalysis, selective adsorption, and separations¹⁻³. Their unique pore structures provide molecular-scale confinement through diverse and intricate pore networks, enabling shape-selective transformations and high-efficiency separation processes⁴⁻⁸. A critical factor underlying these functions is the diffusion of guest molecules within the confined zeolitic framework, which strongly affects reaction kinetics, product selectivity, and catalyst longevity^{7,9,10}. The intricate topology of zeolite frameworks exerts a profound influence on molecular diffusion¹¹. Elucidating how pore geometry and connectivity govern confined diffusion remains a fundamental scientific challenge and a critical prerequisite for the rational design of zeolitic catalysts.

Diffusion in zeolites deviates markedly from that in homogeneous systems due to pronounced channel confinement effects^{12,13}. In these microporous environments, diffusion is not only influenced by guest–guest interactions but also dominated by guest–framework interactions^{14,15}. Conventional predictive methods often rely on empirical comparisons between molecular kinetic diameters and pore sizes, but such approaches offer limited quantitative accuracy, particularly when the guest and pore dimensions are similar¹⁶⁻²⁴. Minor structural variations in the pore environment can lead to diffusivity changes spanning orders of magnitude. While molecular dynamics (MD) simulations can access quantitative diffusion parameters for specific systems, they are often restricted to individual frameworks and lack transferable mechanistic insights across different topologies. This gap hinders a unified understanding of confined diffusion and challenges the predictive design of zeolite-based materials.

Fortunately, machine learning (ML) has emerged as a transformative approach in materials science for uncovering complex structure-property relationships²⁵⁻²⁹. By leveraging large-scale theoretical or experimental datasets, machine learning extracts the quantitative influence of structural features on material properties, shifting traditional empirical workflows toward data-driven, high-precision predictions³⁰⁻³³. This approach enables high-throughput screening of vast material libraries and significantly enhances design optimization and performance tuning^{21,34-37}. In the context of zeolites, ML has been successfully applied to framework classification³⁸, structural screening^{21,39}, synthesis optimization⁴⁰⁻⁴², and catalytic performance prediction^{37,43}. Beyond zeolitic systems, recent studies have also demonstrated the potential of ML for

predicting molecular diffusion under confinement in bulk and mesoporous environments. For example, Leverant *et al.* developed artificial neural network (ANN) models to predict self-diffusion coefficients (D_s) of chemically diverse molecules confined within idealized pore geometries, spanning from small polar species to long-chain hydrocarbons, and applicable to both bulk and confined liquid systems⁴⁴. Despite these advances, ML models specifically designed for predicting zeolite diffusion coefficients remain scarce^{45,46}. In contrast to mesoporous or idealized confined systems, diffusion in zeolites is dominated by strong guest–framework interactions and complex crystalline pore topologies, which pose additional challenges for model transferability and interpretability. As a consequence, several key challenges persist, including the limited availability of high-quality diffusion data and the lack of interpretable descriptors that can capture complex structural features of zeolites, such as channel tortuosity, pore connectivity heterogeneity, and cross-sectional area fluctuations. Overcoming these challenges is crucial to enhancing both the predictive accuracy and mechanistic interpretability of ML models for zeolite diffusion.

To overcome these limitations, we combine high-throughput molecular simulations with interpretable machine learning to establish a structure-diffusion database. We further devise a suite of topological descriptors that quantitatively capture pore tortuosity and connectivity directly from the diffusion mechanism. This approach not only reveals the structural principles governing confined diffusion in zeolites but also enhances the predictive power and physical interpretability of diffusion models, providing a scalable computational platform for the rational design and screening of porous materials.

Results

Construction of diffusion database and machine learning models

The first step toward establishing a unified structure-diffusivity relationship is the construction of comprehensive datasets. Although some scattered data exists, a systematic, large-scale database covering diverse zeolite pore types and providing molecular dynamics-based diffusion coefficients remains unavailable. To this end, we developed a high-throughput computational platform (Fig. 1) for analyzing and predicting diffusivity in zeolitic materials. The platform takes structural files of zeolites (e.g., CIF or PDB

formats) as input and provides an integrated workflow that extracts key structural information, performs adsorption calculations, and conducts molecular dynamics simulations to determine diffusion coefficients. Using this platform, we evaluated the structural features and methane diffusion coefficients of 4,194 zeolite frameworks, drawing their structures from the International Zeolite Association (IZA) database⁴⁷ and the Predicted Crystallography Open Database (PCOD)⁴⁸. This effort yields an initial dataset underpinning a systematic exploration of structure-diffusion relationships in zeolites.

The resulting dataset comprises key structural parameters, including pore-limiting diameter (PLD), largest cavity diameter (LCD), global cavity diameter (GCD), accessible volume fraction (V_{acc}), accessible surface area (ASA), framework density (FDSi), and bulk density (ρ). PLD defines the narrowest constriction that a diffusing molecule must traverse; frameworks with comparable cavity sizes but larger PLD generally facilitate faster diffusion⁴⁹. LCD captures the maximum free volume within channels, where larger cavities mitigate the residence-time penalty associated with molecular trapping in cages. GCD reflects the dominant cavity size of the framework and often correlates with adsorption capacity (Fig. 2a). V_{acc} quantifies the fraction of volume accessible to a probe molecule, with higher values typically implying greater porosity and enhanced molecular mobility. ASA characterizes the internal framework surface that is accessible to a probe molecule, reflecting the extent of surface exposure within the pore network. FDSi, the number of tetrahedral atoms per 1,000 Å³, provides a measure of framework compactness, where high values restrict available diffusion pathways. Finally, the bulk density ρ encapsulates both framework density and channel void fraction, with denser frameworks generally associated with narrower pores and lower diffusivities. These parameters, frequently employed as descriptors in machine learning models for diffusivity prediction, provide a structural basis for exploring structure-diffusivity relationships in porous materials⁵⁰⁻⁵². Complementing these structural metrics, the dataset also includes methane diffusion coefficients obtained through molecular dynamics simulations in the high-throughput computational platform. The resulting diffusion coefficients span a range of 10⁻¹² to 10⁻⁷ m²/s (Fig. 2b), which is consistent with previous reports⁵³. Diffusion is strongly influenced by framework geometry; in our data, when PLD ≤ 5 Å, the diffusion coefficient increases with increasing PLD, indicating a positive correlation (Fig. 2c).

However, this trend becomes less clear for $PLD > 5 \text{ \AA}$, indicating that while diffusion in narrow-pore systems is largely governed by geometric confinement, the effect weakens as pore size increases. This dataset thus establishes a foundation for elucidating the relationship between framework structure and diffusion performance.

Building on this dataset, we aim to establish a quantitative framework that links geometric structure to diffusion performance. However, this structure-property relationship is inherently complex, shaped by the interplay of multiple structural factors and often exhibiting strong nonlinearity, which complicates direct interpretation. Machine learning methods, with their capacity to capture intricate patterns in high-dimensional data, provide an effective means to elucidate and predict such correlations. All seven structural parameters of the zeolite frameworks were used as geometric descriptors to train machine learning models for predicting diffusion coefficients. Although each descriptor exerts a measurable influence on diffusivity, substantial correlations exist among them (Fig. 2d). For example, the Pearson correlation coefficient between GCD and LCD is as high as 1.00, indicating complete linear dependence and functional redundancy. Highly correlated descriptors can reduce model efficiency, increase training complexity, and potentially lower predictive accuracy. To mitigate this, descriptors with Pearson correlation coefficients (r) greater than 0.95 were reduced to a single representative feature. For instance, given the strong correlation between LCD and PLD ($r = 0.95$), PLD was retained as the representative feature. As a result, PLD, FDSi, V_{acc} and ASA were selected as the input descriptors. An ANN was trained on 90% of the structure dataset using cross-validation, with the remaining 10% reserved as an independent test set.

The machine-learning model achieved coefficients of determination (R^2) of 0.78 on the training set and 0.78 on the test set, with approximately 10% of predictions deviating by one to two orders of magnitude, and a few outliers exceeding two orders (Fig. 2e). To interpret the model's predictions, we employed Shapley Additive Explanations (SHAP) for feature importance analysis⁵⁴. The analysis identified PLD as the dominant structural descriptor influencing molecular diffusion in zeolites (Fig. 2f), consistent with previous findings^{52,54,55}. Despite capturing general trends, substantial discrepancies arise in certain frameworks. For instance, the ML-predicted diffusion coefficients for BOF and EWO zeolites differ significantly from MD simulations. For the BOF zeolite, the ML-predicted diffusion coefficient is $3.13 \times$

$10^{-8} \text{ m}^2/\text{s}$ —approximately thirteen times higher than the MD result of $0.25 \times 10^{-8} \text{ m}^2/\text{s}$, indicating a full order-of-magnitude overestimation (Fig. 3a). Specifically, BOF (PLD = 4.61 Å) and PSI (PLD = 4.79 Å) share similar values in key structural parameters, including PLD and FDSi (Table S4). Although BOF and PSI exhibit noticeable differences in both V_{acc} and ASA, the overall impact of these descriptors on diffusion is limited (Fig. 2f). Consequently, the ML model predicts comparable diffusion coefficients for the two zeolites (BOF: $3.13 \times 10^{-8} \text{ m}^2/\text{s}$; PSI: $1.52 \times 10^{-8} \text{ m}^2/\text{s}$). However, in the case of PSI, the ML prediction ($1.52 \times 10^{-8} \text{ m}^2/\text{s}$) aligns more closely with the MD simulation ($8.87 \times 10^{-8} \text{ m}^2/\text{s}$), whereas for BOF, a substantial deviation remains. On the other hand, for the EWO zeolite, the ML-predicted methane diffusivity is $5.15 \times 10^{-10} \text{ m}^2/\text{s}$, which starkly contrasts with the MD-derived value of $1.0 \times 10^{-11} \text{ m}^2/\text{s}$ —a discrepancy of nearly 51-fold (Fig. 3a). Interestingly, NPO—a zeolite with a nearly identical PLD (3.65 Å vs. 3.66 Å for EWO)—exhibits a much higher diffusion coefficient of $2.05 \times 10^{-8} \text{ m}^2/\text{s}$ according to MD simulations. In particular, EWO has a higher accessible pore volume fraction (3.17 %) than NPO (2.47 %), suggesting that its internal void space should, in principle, better facilitate methane diffusion. However, the unexpectedly low diffusion of methane in EWO indicates that local structural features impose additional constraints on molecular motion. These observations underscore the limitations of using only conventional descriptors and highlight the need to incorporate local framework characteristics when predicting diffusion in confined zeolite channels.

To unravel the mechanisms of confined diffusion in zeolites, we further examined the diffusion trajectories, velocity autocorrelation functions (VACFs), and free-energy landscapes of methane as a representative guest molecule. Six representative frameworks were selected and classified into two groups according to their PLDs: PSI, JRY and BOF, with relatively large PLDs of 4.34–4.79 Å; and NPO, EWO and a hypothetical framework (ID: 8212223, hereafter PCOD-8212223), with smaller PLDs of 3.65–3.66 Å. Although PLDs are similar within each group, the diffusion behaviors vary significantly, indicating that factors beyond PLD influence molecular mobility. First, to gain microscopic insight into molecular motion, MD trajectories were analyzed⁵⁶. In the large-PLD group (PSI, JRY and BOF), the diffusion coefficients of methane in these three zeolites span more than an order of magnitude. In PSI, where the channels are straight, the diffusion pathway is nearly linear (Fig. S9a), with the methane molecule moving smoothly along the

channel with fewer collisions with the framework. Directional reversals are rare, indicating an unobstructed transport environment. In contrast, methane motion within the tortuous channels of JRY (Fig. S9b) exhibits tortuosity, suggesting partial hindrance from the channel topology. This effect is even more pronounced in BOF, whose highly tortuous channels produce complex diffusion dynamics. The methane trajectory shows frequent bending and directional changes (Fig. 3d), with overlapping path segments colored in blue (early time) and red (later time), indicating repeated collisions and back-and-forth movement within the same region. This observation is consistent with previous findings⁵⁶, highlighting the critical role of channel morphology in regulating molecular diffusion. On the other hand, methane diffusion in NPO remains linear but the trajectory is more spatially concentrated than that in PSI, primarily due to the smaller PLD of NPO (Fig. 3d). However, in EWO zeolites, the diffusion trajectory of a methane molecule reveals a “cluster-like” pattern, indicating confinement within specific regions situated in the wider sections of the channels in EWO, where referred to as “cavity-in-channel”. The channels between adjacent cavities are relatively narrow, allowing methane molecules to migrate to neighboring cavities only occasionally, driven by thermal fluctuations that provide sufficient energy, before residing in the new cavity for a duration. This diffusion behavior contrasts sharply with the linear diffusion observed in NPO and PSI, highlighting the significant influence of the internal structure of the zeolite on the diffusion behavior of methane molecules.

To elucidate how zeolite channel structures modulate diffusion, we constructed two-dimensional FESs projected along the diffusion pathways to quantify the associated energy barriers (Fig. 4a–f). Lower barriers correspond to enhanced diffusion, whereas higher barriers impede transport⁵⁷. Complementary VACF analysis was employed to probe collisions between methane molecules and the zeolite framework, providing additional insights into the diffusive dynamics^{56,58–60}. Shorter velocity relaxation times indicate more frequent collisions and slower diffusion, whereas broader positive regions in the VACF reflect faster transport, and pronounced negative peaks signify momentum reversals indicative of hindered diffusion. In the first group of zeolites (PSI, JRY and BOF), the two-dimensional FESs (Fig. 4a–c) exhibit fluctuations below $3 k_B T$, indicating that channel bending alone does not impose significant energetic barriers. VACF analysis (Fig. S8a) further distinguishes their diffusive behavior: PSI displays the slowest decay and shallowest negative peak, reflecting minimal collisions and highly efficient diffusion along straight

channels; JRY exhibits a moderate negative peak, suggesting that slight tortuosity imposes some confinement; and BOF shows the most pronounced negative peak, consistent with frequent collisions and substantial momentum loss in its highly tortuous channels. Together, these observations suggest that diffusion in this group is mainly limited by collision dynamics rather than energetic barriers, with BOF exhibiting slower transport due to its complex channel geometry. These findings are consistent with our previous work⁵⁶, underscoring the critical role of channel topology in governing molecular diffusion.

In contrast, the diffusion-limiting mechanisms in the second group of zeolites (PCOD-8212223 and EWO) are fundamentally different. Methane VACFs in this group exhibit significant fluctuations, complicating direct interpretation of collision dynamics (Fig. S11a). Here, FES analysis provides a more reliable assessment: NPO shows a relatively smooth energy landscape with a diffusion barrier of $\sim 5.2 k_B T$ (Fig. 4d), indicating moderate resistance to transport, whereas EWO exhibits a substantially higher barrier of $\sim 9.5 k_B T$ (Fig. 4f), nearly twice that of NPO, and consistent with markedly hindered diffusion. Structural analysis reveals the origin of this disparity: NPO displays minor variations along the diffusion pathway, with an LCD of 4.17 Å, only 1.14 times its PLD; in EWO, the LCD reaches 5.21 Å ($1.42 \times \text{PLD}$), and the shortest distance between cavity and bottleneck is approximately 2 Å. This abrupt local constriction creates steep free-energy barriers, prolonging methane residence within cavities and severely restricting overall transport.

Based on the analysis of the diffusion behavior described above, the confined diffusion of molecules in zeolites can be categorized into three types (Fig. 4g). In straight channels, gas molecules move almost unhindered, similar to race cars accelerating along a straight track, resulting in the fastest diffusion. In tortuous channels, continuous changes in direction increase the likelihood of collisions with the framework, akin to cars navigating a winding track, which restricts their mobility. In channels with cross-sectional fluctuations, molecules must overcome higher energy barriers at local bottlenecks, forcing them to slow down, analogous to vehicles proceeding cautiously along a narrow road, thereby markedly reducing the overall diffusion rate. However, conventional geometric descriptors fail to capture channel tortuosity and local cross-sectional variations, making it difficult to fully rationalize the underlying diffusion mechanisms and underscoring the need for more refined topological analyses.

Enhanced machine learning model incorporating physics-informed structural descriptors

The limited predictive accuracy of the previously described machine learning model primarily stems from its inability to capture essential aspects of the diffusion mechanism. Specifically, the conventional structural descriptors employed—PLD, LCD, FDSi, V_{acc} , and ASA partially capture the geometric features of zeolite frameworks but are insufficient to reflect finer structural features such as channel tortuosity and variations in pore cross-section, both of which play a critical role in governing molecular diffusion. Since feature selection largely determines the accuracy and generalizability of machine learning models, addressing this limitation is critical. To this end, we define a set of geometrical descriptors derived from an energy grid-based approach that integrate physical interpretability with computational efficiency and universality (Fig. 5a, 5b). These descriptors capture key structural features of zeolite channels, including channel tortuosity (τ ; Fig. S19); a set of cross-sectional area metrics—the minimum (A_{min}), maximum (A_{max}), median (A_{med}), and mean (A_{mean}) accessible areas—together with their standard deviation (A_{SD}), which quantifies pore-size fluctuations (Fig. S20); and a smoothness parameter defined as the ratio of the pore-limiting diameter to the largest cavity diameter (PLD/LCD)⁶¹. Briefly, τ characterizes channel tortuosity, the area metrics quantify the available space for molecular diffusion and its heterogeneity, A_{SD} measures the severity of cross-sectional variations, and PLD/LCD reflects the smoothness of pore walls. Representative values of these descriptors for different zeolite frameworks, along with corresponding channel morphologies, are shown in Figs. S21 and S22, illustrating their effectiveness in capturing geometric features. Table S4 presents the calculated values of the aforementioned descriptors for one-dimensional zeolite frameworks in the IZA Zeolite Structure Database.

These descriptors were subsequently used as input features to train machine learning models capable of capturing the complex interplay between structure and diffusion. Pearson correlation analysis revealed that descriptors such as PLD/LCD, τ , and A_{SD} are less correlated with other descriptors, exhibiting low redundancy and suggesting that they encode distinct, complementary structural information. Again, to mitigate multicollinearity, highly correlated features ($r > 0.95$) were merged, retaining only the most

representative descriptors. Ultimately, a set of nine non-redundant features (including PLD, PLD/LCD, A_{SD} , and τ) was selected for model training (Fig. 6a).

An ANN was trained incorporating the descriptors introduced in this study. We systematically compared the predictive performance of machine learning models with and without the proposed descriptors. For zeolites where the baseline model already performed well, the enhanced model maintained high accuracy. For systems with less reliable initial predictions, the revised model showed an observable improvement, indicating that the introduced descriptors provide complementary structural information. The enhanced model achieved a test-set R^2 of 0.91 and an RMSE of 0.34 (Fig. 6b), outperforming an ANN trained solely on conventional descriptors (test $R^2 = 0.78$).

To assess algorithm dependence, we further evaluated three widely used tree-based algorithms: Random Forest (RF), LightGBM (LGBM), and Extreme Gradient Boosting (XGBoost), which have been reported to perform well for material properties^{37,51,52,55}. All four models exhibited comparable predictive accuracy on the test set ($R^2 \approx 0.90$ – 0.91 ; RMSE ≈ 0.34 – 0.35 ; Fig. 6b), indicating that the current descriptors successfully capture the dominant structural features governing diffusion. This similarity in performance across models suggests that the predictive accuracy is primarily limited by the information content of the dataset, a conclusion further supported by the learning curves (Figs. S26 and S27), which show that increasing the training data beyond $\sim 50\%$ of the full dataset yields only limited improvements in validation and test performance. The plateau indicates that the feature space is largely saturated, with the principal geometric descriptors already capturing the structural information relevant to methane diffusion. The current dataset, while representing a relatively comprehensive collection, is inherently constrained by the structural diversity of the currently known zeolites. Most frameworks in the IZA and PCOD databases are microporous, with pore-limiting diameters concentrated in the range of 3–8 Å and channel geometries spanning only a limited portion of the topological space. As a result, the descriptor space is sampled within a comparatively narrow structural domain, which may contribute to the saturation behaviour observed in the learning curves. Together, these observations indicate that the present performance bottleneck is governed primarily by the structural diversity of existing zeolite materials. It is worth noting that, despite these constraints, the model achieves high predictive accuracy for zeolite diffusion coefficients ($R^2 = 0.91$).

within this domain, suggesting that the key structure-diffusivity relationships have been effectively captured. In comparison with end-to-end models that learn directly from crystal structures, the designed descriptors may not capture all structural details. However, such approaches typically require very large datasets⁶². By comparison, the descriptors used here offer strong predictive performance with relatively modest dataset sizes while preserving interpretability and physical insight. As the synthesis of zeolites with larger pores and more diverse channel architectures continues to advance, the high-throughput platform can be adapted to incorporate additional frameworks. Future work will focus on expanding the database to include hierarchical and mesoporous materials, thereby broadening the applicability of the developed descriptors and models.

Among the four models, ANN showed the most consistent performance, exhibiting the smallest training-validation gap (Fig. S28) and the lowest fold-to-fold variability (Table S3), and was therefore selected as the primary predictive framework. For representative systems such as BOF and EWO, ANN predictions closely matched MD-derived diffusion coefficients with reduced error (Table S5). Across the full test set, most predicted diffusivities fell within one order of magnitude of reference values (Fig. 6c), demonstrating strong generalizability. Furthermore, the framework also exhibits strong transferability. We performed transfer learning (TL) on 400 ethane diffusion data points by freezing the first two hidden layers of the ANN (Table S5). On an independent test set, the transfer-learned model achieved a higher coefficient of determination ($R^2 = 0.81$) than a model trained from scratch ($R^2 = 0.67$) (Fig. S31). Consistent improvements were observed in cross-validation, with an average R^2 of 0.84 ± 0.079 and RMSE of 0.52 ± 0.108 , compared with $R^2 = 0.77 \pm 0.085$ and RMSE = 0.64 ± 0.146 for the scratch-trained model (Table S6). Similar improvements were obtained for ethene and methanol molecules (Tables S7 and S8, Figs. S32 and S33). These results show that the framework is transferable beyond methane across chemically distinct guest molecules.

SHAP analysis of diffusion mechanisms

To further elucidate the contributions of individual features to diffusion behavior, we employed SHAP analysis⁵⁴, which quantifies the impact of each input feature on model predictions by evaluating changes in

the output with and without that feature across multiple feature combinations. This method provides a consistent framework for interpreting how structural descriptors govern diffusion, both at the level of individual frameworks and across the entire dataset, yielding interpretable insights into feature importance and their complex interdependencies. As shown in Fig. 6e, PLD remained the dominant factor influencing diffusion; however its relative importance declined upon introduction of descriptors proposed in this study. This shift highlights the critical role of structural complexity (particularly as captured by PLD/LCD, A_{SD} , and τ) in governing confined diffusion. Larger pore apertures (higher PLD), more uniform channels (PLD/LCD approaching 1), and greater V_{acc} all favored enhanced molecular mobility. In contrast, greater A_{SD} and tortuosity were consistently associated with reduced diffusivity, in accordance with steric hindrance and increased collision frequency within non-uniform, meandering channels. Interestingly, framework density played a relatively minor role in regulating diffusion (Fig. 6d).

A more detailed SHAP decomposition of channel tortuosity (τ) and the standard deviation of channel cross-sectional area (A_{SD}) further reveals nuanced structure-property relationships (Fig. S29). The SHAP values of τ exhibit a clear negative correlation: in low-tortuosity regions ($\tau \leq 1.1$), values are mostly positive, indicating that relatively straight channels facilitate diffusion; as τ increases, SHAP values become negative, reflecting inhibition of diffusion due to more frequent collisions along curved paths, especially pronounced in small-pore zeolites (PLD < 5 Å) and weaker in larger-pore systems (PLD > 5 Å). Similarly, A_{SD} strongly inhibits diffusion at lower values ($A_{SD} \leq 10$ Å²), with SHAP values decreasing rapidly as A_{SD} rises, while for larger A_{SD} (mostly corresponding to high-PLD zeolites), SHAP values plateau, suggesting that sufficiently wide channels partially buffer the adverse effect of cross-sectional fluctuations. These results indicate that tortuosity and cross-sectional heterogeneity act synergistically with PLD to regulate molecular diffusion in zeolite channels.

At the sample-specific level, individualized SHAP profiles reflect how each descriptor quantitatively influences diffusion in a given framework. For instance, in BOF, PLD remains the most influential descriptor, while tortuosity ranks second due to the pronounced tortuosity of its channels. Conversely, its relatively uniform channel width rendered A_{SD} a slightly positive factor, enhancing overall mobility (Fig.

6f). These individualized attribution profiles provide a mechanistic view of how multiple structural features interact in concert to modulate diffusion.

In summary, the descriptor framework improves the predictive capacity of structure-diffusion models and offers mechanistic insights into the role of topological complexity in confined diffusion. Unlike conventional geometric metrics, these descriptors capture multiscale heterogeneity, pathway tortuosity, and spatial confinement in a physically grounded manner. By integrating automated descriptor extraction with machine learning-based predictions, we established a high-throughput computational platform capable of rapidly and accurately evaluating methane diffusivity directly from zeolite structural features. While the present work focuses on one-dimensional channel systems, future extensions to three-dimensional interconnected pore networks and hierarchical materials (offer broader ranges of pore sizes) will further expand their applicability. Overall, the approach provides a versatile and extensible toolkit for high-throughput screening, property prediction, and the rational design of porous materials with tailored diffusion characteristics.

Discussion

In this study, we developed a high-throughput computational framework that enables physically interpretable prediction of molecular diffusion in zeolites by systematically linking pore topology to diffusion behavior across diverse frameworks. Leveraging this platform, we constructed a relatively comprehensive zeolite diffusion database, encompassing 4,194 distinct frameworks, providing a robust foundation for mechanistic studies of confined diffusion. Systematic analysis revealed that molecular diffusion in zeolites is governed not by a single geometric parameter, but by the combined effects of channel topology—particularly tortuosity, which determines the effective diffusion path length and molecular collision frequency, and spatial fluctuations in cross-sectional area, which introduce variations in free energy barriers along diffusion pathways.

To address the limitations of conventional geometric descriptors, we introduced a set of physically meaningful, topologically informed descriptors, including channel tortuosity, accessible cross-sectional area, and its standard deviation. These descriptors capture structural heterogeneity that directly governs

molecular collisions and free energy barriers along diffusion pathways. Incorporating these descriptors into machine-learning models substantially improves predictive performance ($R^2 > 0.91$ on the test set), demonstrating that topology-aware representations encode essential physical constraints that are missed by traditional descriptors. Quantitative SHAP analysis clarifies the mechanistic contributions of individual features: larger pore size and channel uniformity facilitate diffusion, whereas increased tortuosity and cross-sectional heterogeneity impose significant transport limitations. The observed inhibitory effects are most prominent in small-pore frameworks and progressively weaken as the pore size increases, underscoring a clear structure-dependent and synergistic regulation of transport behavior. Furthermore, the proposed modeling framework exhibits strong transferability. By applying transfer learning from methane to ethane, ethene, and methanol molecules, we demonstrate that the model not only achieves high predictive accuracy but can also be effectively extended to other small organic molecules. This confirms that the same set of descriptors and ANN architecture can reliably predict diffusion properties across different light hydrocarbons, highlighting the versatility and robustness of the approach.

By integrating automated feature extraction with predictive modeling, this work establishes a unified workflow connecting zeolite structure, topology, and diffusion behavior. Beyond diffusivity prediction, the framework provides a foundation for broader investigations of transport phenomena in zeolites. Future work may extend this approach to diffusion-reaction coupling, multicomponent systems, multidimensional channel networks, and experimentally relevant conditions such as molecular loading and temperature. Further methodological advancements—including the integration of machine-learning-based interatomic potentials and the explicit incorporation of physicochemical heterogeneities (e.g., Brønsted acid site distributions and exchanged metal ions)—represent promising directions toward a more realistic description of confined diffusion. Together, these developments could deepen mechanistic insight under realistic conditions and further expand the framework's applicability to the rational design and screening of zeolites and related nanoporous materials.

Methods

Structural database and screening

A high-throughput computational platform was developed to evaluate methane diffusion in zeolite frameworks. Structural data were sourced from two repositories: 256 from the IZA database⁴⁷ and 100,000 from the PCOD⁴⁸. Following geometric screening based on methane kinetic diameter, approximately 4,194 one-dimensional pure-silica frameworks with pore-limiting diameters ≥ 3.5 Å were selected. Further details on the computational screening methodology can be found in Supplementary Dataset and methods.

Adsorption simulation

High-throughput Monte Carlo (MC) simulations were conducted to generate initial configurations for MD. Each zeolite framework was simulated as a supercell with all box lengths exceeding 40 Å, containing 100 methane molecules. Prior to the adsorption simulations, inaccessible cavities and occluded pockets were identified using the blocked-sphere⁶³ procedure implemented in Zeo++. The resulting block files were imported into RASPA to restrict methane sampling exclusively to physically accessible pore regions. Simulations were conducted using the NVT ensemble, with the temperature set at 298 K. The TraPPE-UA⁶⁴ and TraPPE-Zeo⁶⁵ force fields were employed for methane (as well as ethane, ethene and methanol) and framework atoms, respectively, with Lorentz-Berthelot mixing rules used for cross-interactions. Additional simulation parameters are detailed in Table S1.

Molecular dynamics simulations

MD simulations were carried out using the structures obtained from MC simulations to extract methane self-diffusion coefficients. All simulations were performed under the infinite-dilution limit, averaging the mobility of 100 methane molecules while neglecting methane–methane interactions. The same process was also applied to other molecules (e.g. ethane, ethene, and methanol). Simulations were conducted in the canonical (NVT) ensemble at 298 K, with temperature controlled via a Nosé–Hoover thermostat. The leapfrog Verlet algorithm was used for integration. A Lennard–Jones cutoff of 14 Å was applied to all non-bonded interactions. MD simulations were performed with a time step of 1 fs. Each system was equilibrated

for 10^6 steps, followed by 2×10^7 production steps. Trajectories were sampled every 1,000 steps, and three independent runs were performed per framework.

Descriptor development

To capture the geometric complexity of zeolite channels and their influence on methane diffusion, we developed a set of topology-aware structural descriptors based on three-dimensional Lennard-Jones (LJ) potential energy landscapes. Each framework was discretized into a $0.2 \text{ \AA}^{66}$ resolution grid, and the potential energy experienced by a methane molecule was calculated at each grid point. Regions with potential energies below 300 kJ/mol were identified as accessible volumes.

From these accessible domains, diffusion pathways were reconstructed and characterized by several geometric descriptors. Tortuosity was defined as the ratio between the arc length of the reconstructed pathway and the straight-line distance. Cross-sectional variations along the path were quantified by statistical measures of accessible area, including $A_{\max}$, $A_{\min}$, A_{med} , A_{mean} , and A_{SD} . Additionally, the PLD/LCD ratio was incorporated as a measure of pore wall roughness⁶¹. Further details and illustrative examples are provided in Figs. S12-18. Additional methodological details are provided in the Supplementary Information.

Machine learning

The dataset was partitioned into a training and validation set (90%) and an independent test set (10%), with the latter reserved exclusively for the final evaluation of model generalization performance. Hyperparameter optimization and model training were performed using the training/validation set under a cross-validation framework. Four supervised machine learning algorithms were considered: RF, LGBM, XGBoost, and ANN. The RF, LGBM, and XGBoost models were implemented using the scikit-learn library, while the ANN was developed using PyTorch. Model training and hyperparameter optimization were conducted using 5-fold cross-validation. Hyperparameters were tuned via random search over predefined parameter distributions. The optimal hyperparameter values for each model are summarized in Table S2. Using these optimal settings, the final models were retrained and evaluated using 10-fold cross-validation. Additional methodological details are provided in the Supplementary Information.

Full algorithm details

Full details of the machine learning algorithms and hyperparameter settings are provided in the Supplementary Information.

Data availability

All data supporting the findings of this study are available in the main text and the Supplementary Information. The generated zeolite structural descriptors and diffusion datasets are publicly available at <https://doi.org/10.5281/zenodo.18858951>⁶⁷. Source data are provided with this paper. The zeolite framework structures used in this study were obtained from the IZA Zeolite Structure Database (<https://www.iza-structure.org/databases>) and the Predicted Crystallography Open Database (PCOD, <https://mwdeem.org/PCOD>).

Code availability

The code repository is stored at <https://doi.org/10.5281/zenodo.18858951>⁶⁷ and https://github.com/WangXiaobao-j/ZeoDiff_HTP.

References

1. Davis, M. E. Ordered porous materials for emerging applications. *Nature* **417**, 813–821 (2002).
2. Rangnekar, N., Mittal, N., Elyassi, B., Caro, J., Tsapatsis, M. Zeolite membranes – a review and comparison with MOFs. *Chem. Soc. Rev.* **44**, 7128–7154 (2015).
3. Gascon, J. et al. Practical approach to zeolitic membranes and coatings: State of the art, opportunities, barriers, and future perspectives. *Chem. Mater.* **24**, 2829–2844 (2012).
4. Dai, H. et al. Finned zeolite catalysts. *Nat. Mater.* **19**, 1074–1080 (2020).
5. Chen, L.-H. et al. Hierarchically structured zeolites: From design to application. *Chem. Rev.* **120**, 11194–11294 (2020).
6. Gao, S. et al. Cavity-controlled diffusion in 8-membered ring molecular sieve catalysts for shape selective strategy. *J. Catal.* **377**, 51–62 (2019).
7. Fu, D. et al. Unravelling channel structure–diffusivity relationships in zeolite ZSM-5 at the single-molecule level. *Angew. Chem. Int. Ed.* **61**, e202114388 (2022).
8. Fu, D. et al. Elucidating zeolite channel geometry–reaction intermediate relationships for the methanol-to-hydrocarbon process. *Angew. Chem. Int. Ed.* **59**, 20024–20030 (2020).
9. Schneider, D., Mehlhorn, D., Zeigermann, P., Kärger, J., Valiullin, R. Transport properties of hierarchical

- micro-mesoporous materials. *Chem. Soc. Rev.* **45**, 3439–3467 (2016).
10. Kärger, J., Ruthven, D. M. Diffusion in nanoporous materials: Fundamental principles, insights and challenges. *New J. Chem.* **40**, 4027–4048 (2016).
 11. Sun, M.-H. et al. Applications of hierarchically structured porous materials from energy storage and conversion, catalysis, photocatalysis, adsorption, separation, and sensing to biomedicine. *Chem. Soc. Rev.* **45**, 3479–3563 (2016).
 12. Chai, Y., Dai, W., Wu, G., Guan, N., Li, L. Confinement in a zeolite and zeolite catalysis. *Acc. Chem. Res.* **54**, 2894–2904 (2021).
 13. García, E. J., Pérez-Pellitero, J., Jallut, C., Pirngruber, G. D. Modeling adsorption properties on the basis of microscopic, molecular, and structural descriptors for nonpolar adsorbents. *Langmuir* **29**, 9398–9409 (2013).
 14. Bukowski, B. C. et al. Connecting theory and simulation with experiment for the study of diffusion in nanoporous solids. *Adsorption* **27**, 683–760 (2021).
 15. Smit, B. Molecular simulations of zeolites: Adsorption, diffusion, and shape selectivity. *Chem. Rev.* **108**, 4125–4184 (2008).
 16. Cao, P., Zheng, W., Sun, W., Zhao, L. The shape selectivity of zeolites in isobutane alkylation: An investigation using CBMC and MD simulations. *Chem. Eng. Sci.* **245**, 116966 (2021).
 17. Li, B. et al. Understanding the structural properties of zeolites for isobutane alkylation based on adsorption/diffusion behaviors. *Microporous Mesoporous Mater.* **341**, 112040 (2022).
 18. Zhao, Q.-C., Chen, L., Ma, S., Liu, Z.-P. Data-driven discovery of Pt single atom embedded germanosilicate mfi zeolite catalysts for propane dehydrogenation. *Nat. Commun.* **16**, 3720 (2025).
 19. Chizallet, C., Bouchy, C., Larmier, K., Pirngruber, G. Molecular views on mechanisms of brønsted acid-catalyzed reactions in zeolites. *Chem. Rev.* **123**, 6107–6196 (2023).
 20. Tan, J. et al. Ab initio selection of zeolite topology for biomass conversion to para-xylene based on transition state selectivity. *Microporous Mesoporous Mater.* **365**, 112887 (2024).
 21. Kim, J., Abouelnasr, M., Lin, L.-C., Smit, B. Large-scale screening of zeolite structures for CO₂ membrane separations. *J. Am. Chem. Soc.* **135**, 7545–7552 (2013).
 22. Hewitt, D., Pope, T., Sarwar, M., Turrina, A., Slater, B. Machine learning accelerated high-throughput screening of zeolites for the selective adsorption of xylene isomers. *Chem. Sci.* **13**, 13178–13186 (2022).
 23. Li, J.-R., Kuppler, R. J., Zhou, H.-C. Selective gas adsorption and separation in metal-organic frameworks. *Chem. Soc. Rev.* **38**, 1477–1504 (2009).
 24. Jae, J. et al. Investigation into the shape selectivity of zeolite catalysts for biomass conversion. *J. Catal.* **279**, 257–268 (2011).
 25. Kouraytem, N., Li, X., Tan, W., Kappes, B., Spear, A. D. Modeling process-structure-property relationships in metal additive manufacturing: A review on physics-driven versus data-driven approaches. *J. Phys.: Mater.* **4**, 032002 (2021).
 26. Pilania, G., Wang, C., Jiang, X., Rajasekaran, S., Ramprasad, R. Accelerating materials property predictions using machine learning. *Sci. Rep.* **3**, 2810 (2013).
 27. Wei, J. et al. Machine learning in materials science. *InfoMat* **1**, 338–358 (2019).
 28. Wu, W. et al. Quantitative structure-property relationship (qspr) modeling of drug-loaded polymeric

- micelles via genetic function approximation. *PLoS One* **10**, e0119575 (2015).
29. Ren, E., Coudert, F.-X. Prediction of the diffusion coefficient through machine learning based on transition-state theory descriptors. *J. Phys. Chem. C* **128**, 6917–6926 (2024).
 30. Nørskov, J. K., Bligaard, T., Rossmeisl, J., Christensen, C. H. Towards the computational design of solid catalysts. *Nat. Chem.* **1**, 37–46 (2009).
 31. Ulissi, Z. W., Medford, A. J., Bligaard, T., Nørskov, J. K. To address surface reaction network complexity using scaling relations machine learning and dft calculations. *Nat. Commun.* **8**, 14621 (2017).
 32. Hautier, G., Fischer, C. C., Jain, A., Mueller, T., Ceder, G. Finding nature's missing ternary oxide compounds using machine learning and density functional theory. *Chem. Mater.* **22**, 3762–3767 (2010).
 33. Pankajakshan, P. et al. Machine learning and statistical analysis for materials science: Stability and transferability of fingerprint descriptors and chemical insights. *Chem. Mater.* **29**, 4190–4201 (2017).
 34. Orhan, I. B., Daglar, H., Keskin, S., Le, T. C., Babarao, R. Prediction of O₂/N₂ selectivity in metal-organic frameworks via high-throughput computational screening and machine learning. *ACS Appl. Mater. Interfaces* **14**, 736–749 (2022).
 35. Qiao, Z., Peng, C., Zhou, J., Jiang, J. High-throughput computational screening of 137953 metal-organic frameworks for membrane separation of a CO₂/N₂/CH₄ mixture. *J. Mater. Chem. A* **4**, 15904–15912 (2016).
 36. Budhathoki, S., Ajayi, O., Steckel, J. A., Wilmer, C. E. High-throughput computational prediction of the cost of carbon capture using mixed matrix membranes. *Energy Environ. Sci.* **12**, 1255–1264 (2019).
 37. Rzepa, C. et al. Elucidating thermodynamically driven structure-property relations for zeolite adsorption using neural networks. *JACS Au* **4**, 4673–4690 (2024).
 38. Carr, D. A., Lach-hab, M., Yang, S., Vaisman, I. I., Blaisten-Barojas, E. Machine learning approach for structure-based zeolite classification. *Microporous Mesoporous Mater.* **117**, 339–349 (2009).
 39. Erlebach, A., Nachtigall, P., Grajciar, L. Accurate large-scale simulations of siliceous zeolites by neural network potentials. *npj Comput. Mater.* **8**, 174 (2022).
 40. Muraoka, K., Sada, Y., Miyazaki, D., Chaikittisilp, W., Okubo, T. Linking synthesis and structure descriptors from a large collection of synthetic records of zeolite materials. *Nat. Commun.* **10**, 4459 (2019).
 41. Schwalbe-Koda, D. et al. A priori control of zeolite phase competition and intergrowth with high-throughput simulations. *Science* **374**, 308–315 (2021).
 42. Li, X. et al. Machine learning-assisted crystal engineering of a zeolite. *Nat. Commun.* **14**, 3152 (2023).
 43. Ducamp, M., Coudert, F.-X. Prediction of thermal properties of zeolites through machine learning. *J. Phys. Chem. C* **126**, 1651–1660 (2022).
 44. Leverant, C. J., Greathouse, J. A., Harvey, J. A., Alam, T. M. Machine learning predictions of simulated self-diffusion coefficients for bulk and confined pure liquids. *J. Chem. Theory Comput.* **19**, 3054–3062 (2023).
 45. Mace, A., Barthel, S., Smit, B. Automated multiscale approach to predict self-diffusion from a potential energy field. *J. Chem. Theory Comput.* **15**, 2127–2141 (2019).
 46. Ping, J., Muraoka, K., Liu, Z., Nakayama, A. Unveiling the structural factors governing the diffusion of ethene in small-pore zeolites through machine learning. *J. Phys. Chem. Lett.* **16**, 11673–11682 (2025).
 47. International Zeolite Association (IZA), Database of Zeolite Structures, <https://www.iza->

- [structure.org/databases](https://www.rsc.org/structure.org/databases). (accessed 30 August 2024).
48. Pophale, R., Cheeseman, P. A., Deem, M. W. A database of new zeolite-like materials. *Phys. Chem. Chem. Phys.* **13**, 12407–12412 (2011).
 49. Hung, T.-H., Lyu, Q., Lin, L.-C., Kang, D.-Y. Transport-relevant pore limiting diameter for molecular separations in metal–organic framework membranes. *J. Phys. Chem. C* **125**, 20416–20425 (2021).
 50. Lin, J. et al. Machine learning accelerates the investigation of targeted MOFs: Performance prediction, rational design and intelligent synthesis. *Nano Today* **49**, 101802 (2023).
 51. Wang, Y., Jiang, Z.-J., Lu, W., Li, D. Machine learning-assisted exploration of chemical space of MOF-5 analogs for enhanced C₂H₆/C₂H₄ separation. *Angew. Chem. Int. Ed.* **n/a**, e202500783 (2025).
 52. Guo, S. et al. Interpretable machine-learning and big data mining to predict gas diffusivity in metal-organic frameworks. *Adv. Sci.* **10**, 2301461 (2023).
 53. Schuring, D. Diffusion in zeolites: Towards a microscopic understanding. PhD Thesis, Department of Chemical Engineering and Chemistry, Eindhoven University of Technology (2002).
 54. Lundberg, S. M., Lee, S.-I. A unified approach to interpreting model predictions. In *Proc. 31st Int. Conf. Neural Inf. Process. Syst.*, Curran Associates Inc. (2017).
 55. Tan, H., Teng, Y., Shan, G. High throughput computational screening and interpretable machine learning for iodine capture of metal-organic frameworks. *npj Comput. Mater.* **11**, 115 (2025).
 56. Liu, Z. et al. Dependence of zeolite topology on alkane diffusion inside diverse channels. *AIChE J.* **66**, e16269 (2020).
 57. Liu, Z. et al. Molecular self-gating inside a zeolite catalyst. *J. Am. Chem. Soc.* **147**, 6126–6136 (2025).
 58. Voloshin, V. P., Naberukhin, Y. I. Interpreting autocorrelation functions of translational motion of water molecules in molecular dynamics models. *J. Struct. Chem.* **65**, 1050–1057 (2024).
 59. Weber, S. C., Thompson, M. A., Moerner, W. E., Spakowitz, A. J., Theriot, J. A. Analytical tools to distinguish the effects of localization error, confinement, and medium elasticity on the velocity autocorrelation function. *Biophys. J.* **102**, 2443–2450 (2012).
 60. Balucani, U., Brodholt, J. P., Vallauri, R. Analysis of the velocity autocorrelation function of water. *J. Phys.: Condens. Matter* **8**, 6139 (1996).
 61. Foster, M. D., Rivin, I., Treacy, M. M. J., Delgado Friedrichs, O. A geometric solution to the largest-free-sphere problem in zeolite frameworks. *Microporous Mesoporous Mater.* **90**, 32–38 (2006).
 62. Cui, J. et al. Direct prediction of gas adsorption via spatial atom interaction learning. *Nat. Commun.* **14**, 7043 (2023).
 63. Gómez-Álvarez, P., Ruiz-Salvador, A. R., Hamad, S., Calero, S. Importance of blocking inaccessible voids on modeling zeolite adsorption: Revisited. *J. Phys. Chem. C* **121**, 4462–4470 (2017).
 64. Wick, C. D., Martin, M. G., Siepmann, J. I. Transferable potentials for phase equilibria. 4. United-atom description of linear and branched alkenes and alkylbenzenes. *J. Phys. Chem. B* **104**, 8008–8016 (2000).
 65. Bai, P., Tsapatsis, M., Siepmann, J. I. Trappe-zeo: Transferable potentials for phase equilibria force field for all-silica zeolites. *J. Phys. Chem. C* **117**, 24375–24387 (2013).
 66. Mace, A., Barthel, S., Smit, B. Automated multiscale approach to predict self-diffusion from a potential energy field. *J. Chem. Theory Comput.* **15**, 2127–2141 (2019).
 67. Wang, X. et al. Harnessing confinement effect and interpretable machine learning to predict alkane

diffusion in zeolite catalysts. Zenodo, <https://doi.org/10.5281/zenodo.18858951> (2026).

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Author contributions

J.Q., Z.L. and A.Z. designed the research; X.W. carried out the theoretical calculations, collated and analyzed the data, and wrote the paper; M.W., F.L., J.Y., provided helpful discussion; J.Q., Z.L. and A.Z. supervised the work and revised the article. All authors discussed the results and commented on the manuscript.

Competing interests

All authors declare no competing interests.

Figure Legends/Captions

Fig. 1 | Schematic workflow of the high-throughput platform for predicting diffusion performance in zeolite frameworks.

Fig. 2 | Dataset overview and analysis. **a**, Schematic illustration of key pore-structure parameters: PLD, LCD, and GCD. **b**, Distribution of methane diffusion coefficients for the zeolite dataset. **c**, Diffusion coefficients of methane in zeolites as a function of PLD. **d**, Correlation matrix depicting the relationships among topological descriptors in zeolite frameworks. **e**, Comparison between diffusion coefficients obtained from MD simulations and those predicted by the machine-learning model. **f**, Global feature importance ranking derived from mean absolute SHAP values, presented as a bar chart.

Fig. 3 | Schematic representation of the relationship between the topology of zeolites and diffusion coefficients. **a**, Comparison of diffusion coefficients for PSI, JRY, and BOF obtained from MD simulations and ML predictions. **b**, Comparison of diffusion coefficients for NPO, PCOD-8212223, and EWO obtained from MD simulations and ML predictions. **c**, Channel structure of NPO, BOF and EWO along the z , x , and z directions, respectively (color code: channel, sky-blue; framework, gray). **d**, Trajectory of one methane inside NPO, BOF and EWO in z , x and z directions, respectively. The trajectories are colored with time proceeding (red-white-blue).

Fig. 4 | Mechanistic analysis of the influence of zeolite topology on diffusion coefficients. **a-f**, One-dimensional minimum free-energy pathways extracted from two-dimensional free-energy maps of PSI (a), JRY (b), BOF (c), NPO (d), PCOD-8212223 (e), EWO (f). **g**, Schematic representation of molecular diffusion in zeolite channels, illustrated through a racing-car analogy to depict transport along straight channels, tortuous channels, and channels with fluctuating cross-sectional areas.

Fig. 5 | High-resolution reconstruction of zeolite channel topology via energy-threshold clustering. **a**, Framework structure of BOF, alongside the pore topology reconstructed using an interaction-energy threshold ($E_p = 300$ kJ/mol). **b**, Extraction of channel descriptors from the reconstructed BOF channel network. **c**, Diffusion coefficients of zeolites as a function of A_{SD} . **d**, Diffusion coefficients of zeolites as a function of tortuosity. **e**, Diffusion coefficients of zeolites as a function of PLD/LCD.

Fig. 6 | Machine learning-based prediction and interpretation of diffusion in zeolite frameworks. **a**, Correlation matrix of topological descriptors in zeolite frameworks. **b**, Performance of the four ML algorithms on the test set, with D_s as the target value. **c**, Comparison between diffusion coefficients obtained

from MD simulations and predicted by the ML model. **d**, SHAP summary plot illustrating contributions of various descriptors to the diffusion coefficient prediction. The horizontal axis represents the directional influence of the descriptors on the normalized diffusion coefficient ($f(x)$). Positive SHAP values indicate enhanced diffusion, while negative SHAP values indicate suppressed diffusion. **e**, Global feature importance ranking based on mean absolute SHAP values, shown as a bar chart. **f**, SHAP waterfall plot decomposing the contributions of topological descriptors to the predicted $f(x)$ for the BOF framework. The baseline value ($E[f(x)]$) represents the mean value of the training set, with subsequent modifications made by descriptor-specific contributions, leading to the final model output $f(x)$. The true diffusion coefficients are derived by denormalizing: $f_{\text{true}}(x) = f(x)\sigma_D + \mu_D$. where σ_D and μ_D represent the standard deviation and mean of the diffusion coefficients from the training set, respectively.

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Fig. 1 | Schematic workflow of the high-throughput platform for predicting zeolites diffusion performance.

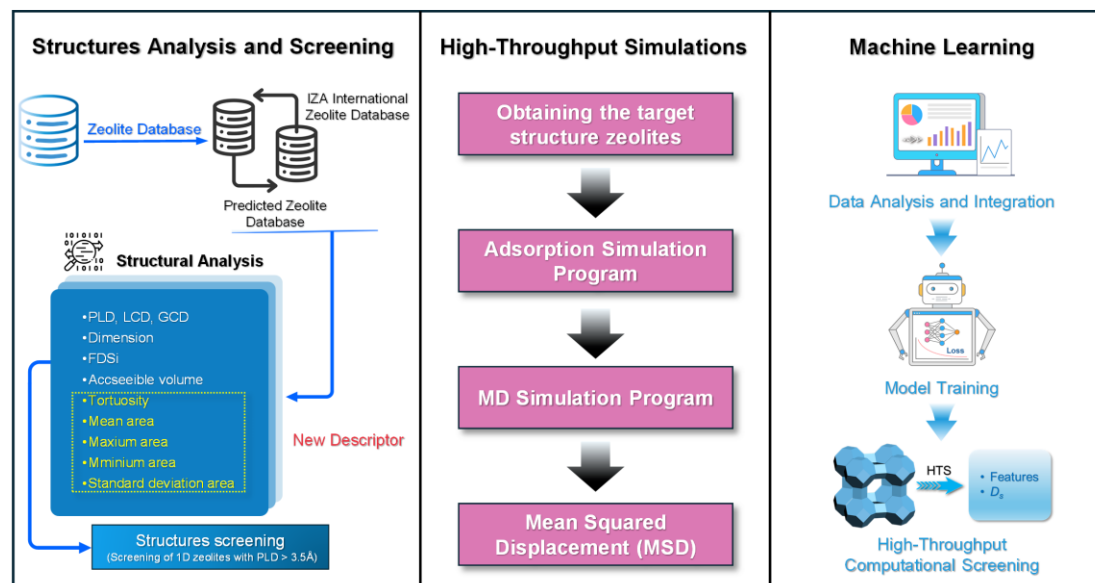


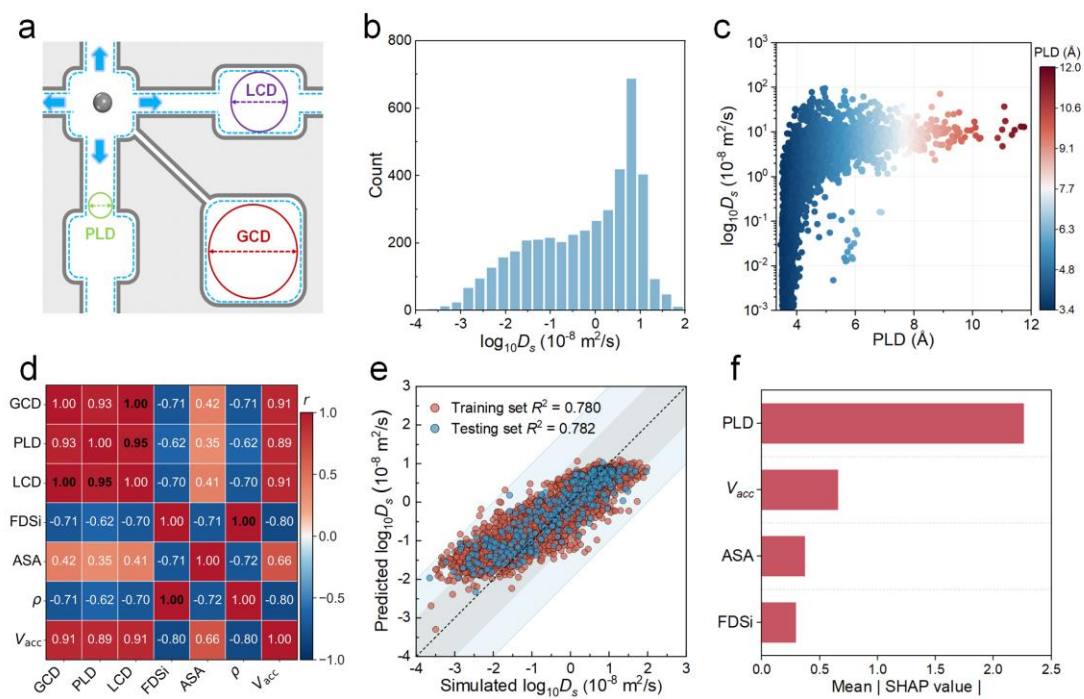
Fig. 2 | Dataset overview and analysis.

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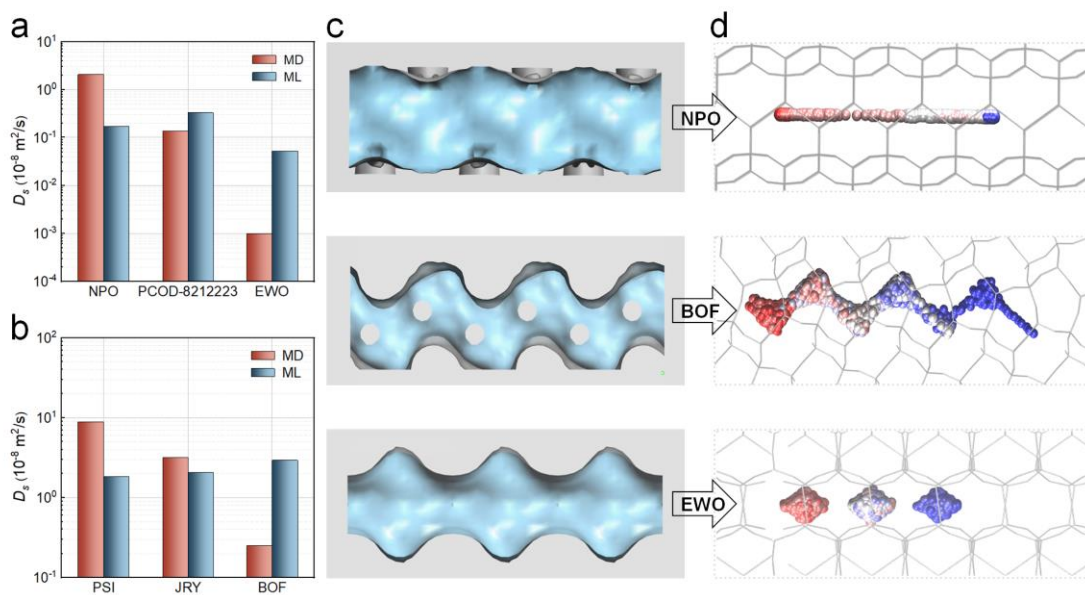


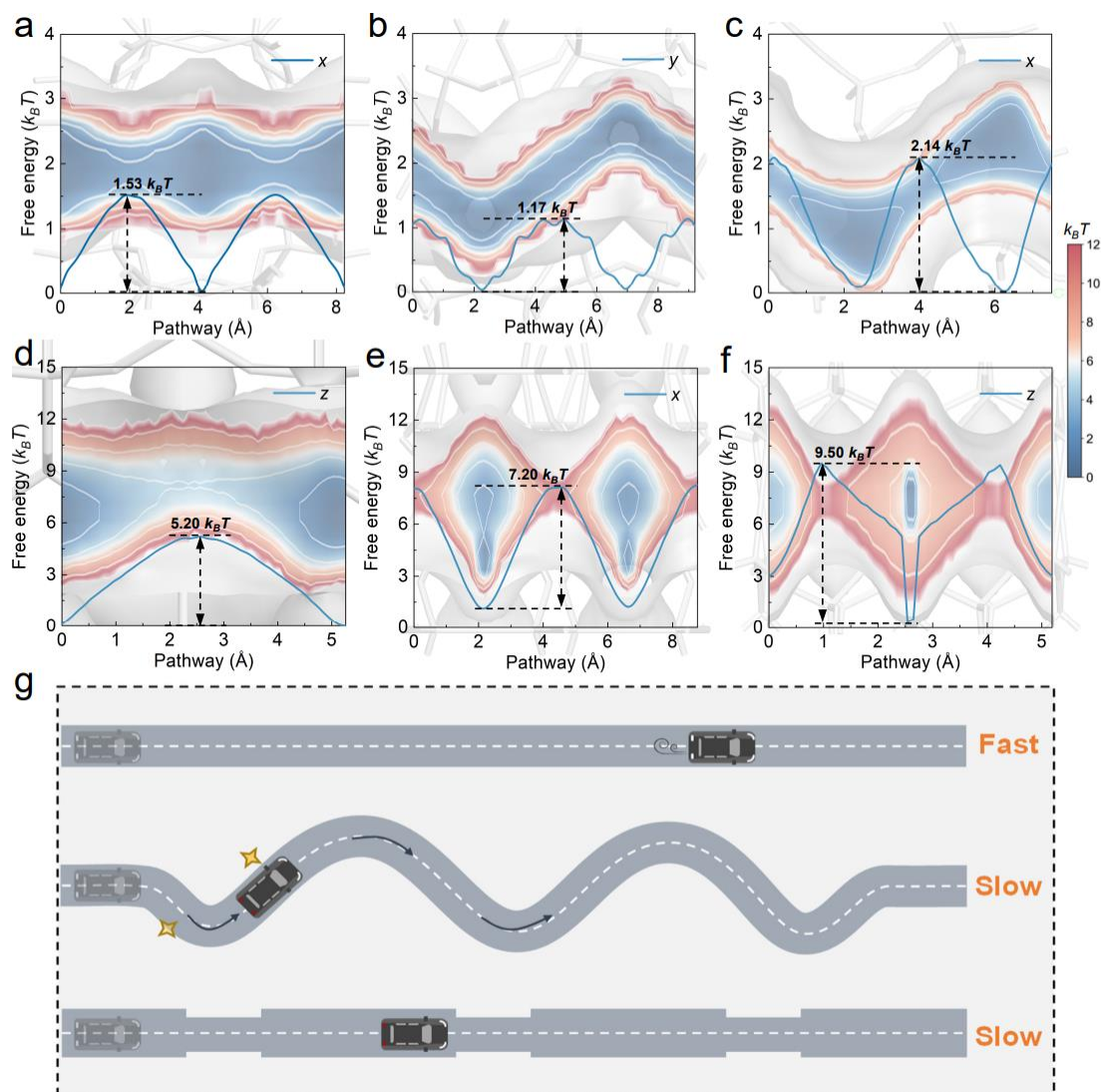
Fig. 4 | Mechanistic analysis of the influence of zeolite topology on diffusion coefficients.

Fig. 5 | High-resolution reconstruction of zeolite channel topology via energy-threshold clustering.

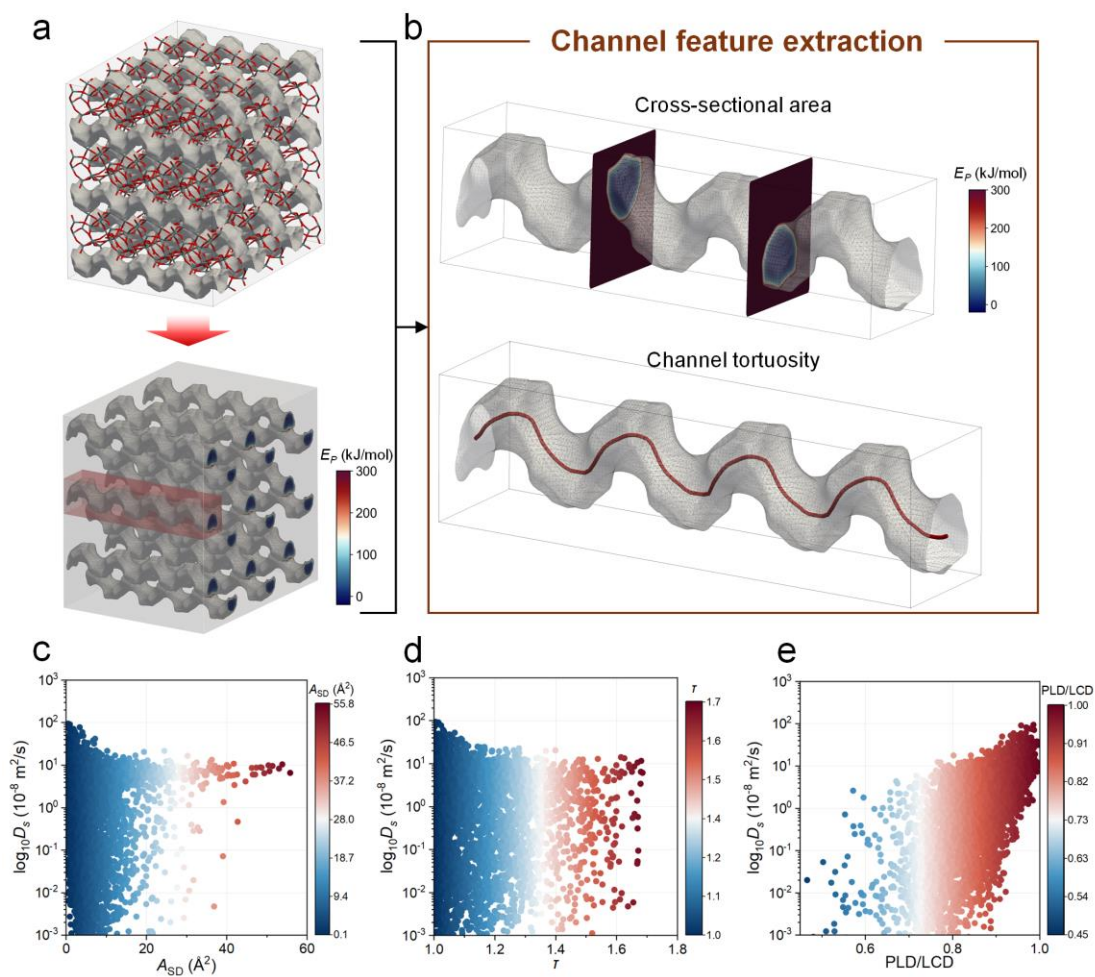
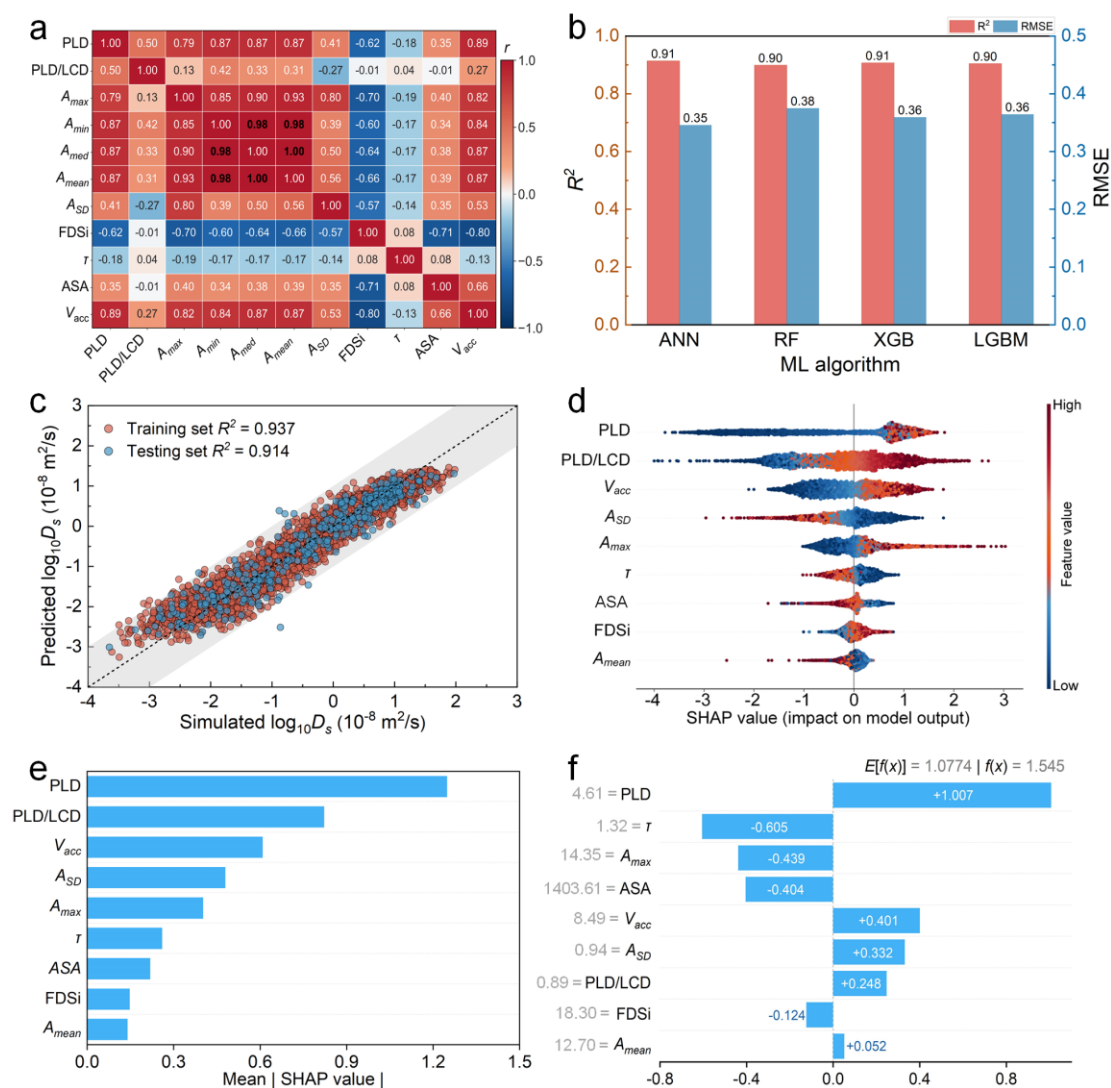


Fig. 6 | Machine learning-based prediction and interpretation of diffusion in zeolite frameworks.

Editorial Summary

This work integrates high-throughput simulations, new diffusion descriptors, and interpretable machine learning to analyze ~4,000 zeolites for predicting molecular diffusion, offering a framework for rational catalyst design.

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